IT5041
Full Stack Software Development with Python

Assignment 1
Creating Real Desktop Software from WxPython using Inno Setup Compiler

Submitted by

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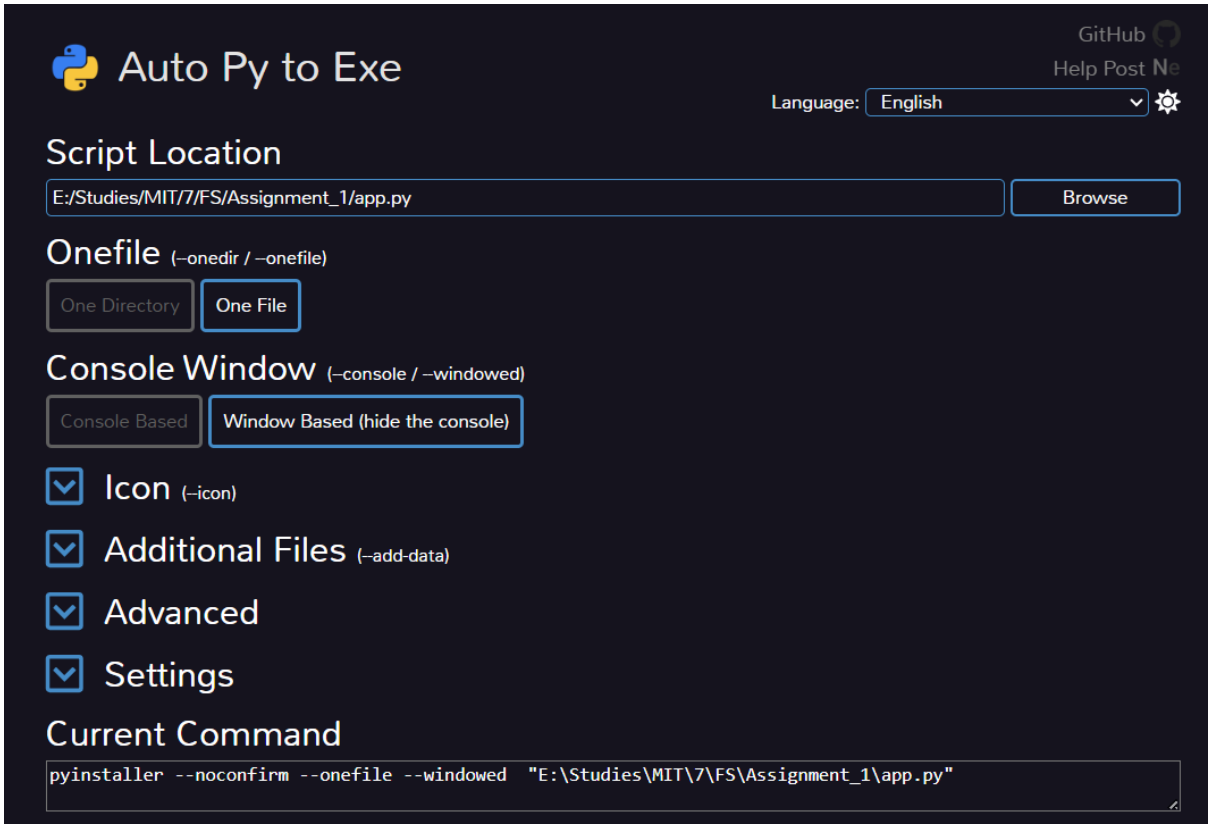
Basic Arithmetic Calculator Application

```
import wx

class ArithmeticApp(wx.Frame):
    def __init__(self, parent, title):
        super(ArithmeticApp, self).__init__(parent, title=title, size=(400, 300))

        panel = wx.Panel(self)

        wx.StaticText(panel, label="Basic Calculator", pos=(90, 20))

        wx.StaticText(panel, label="First Number:", pos=(30, 60))
        self.num1_text = wx.TextCtrl(panel, pos=(180, 55), size=(150, -1))

        wx.StaticText(panel, label="Second Number:", pos=(30, 100))
        self.num2_text = wx.TextCtrl(panel, pos=(180, 95), size=(150, -1))

        wx.StaticText(panel, label="Result:", pos=(30, 140))
        self.result_text = wx.TextCtrl(panel, pos=(180, 135), size=(150, -1),
style=wx.TE_READONLY)

        btn_add = wx.Button(panel, label="+", pos=(50, 190))
        btn_sub = wx.Button(panel, label="-", pos=(110, 190))
        btn_mul = wx.Button(panel, label="×", pos=(170, 190))
        btn_div = wx.Button(panel, label="÷", pos=(230, 190))
        btn_mod = wx.Button(panel, label="%", pos=(290, 190))

        btn_add.Bind(wx.EVT_BUTTON, self.on_add)
        btn_sub.Bind(wx.EVT_BUTTON, self.on_subtract)
        btn_mul.Bind(wx.EVT_BUTTON, self.on_multiply)
        btn_div.Bind(wx.EVT_BUTTON, self.on_divide)
        btn_mod.Bind(wx.EVT_BUTTON, self.on_modulus)

        self.Centre()
        self.Show()

    def get_inputs(self):
        try:
            num1 = float(self.num1_text.GetValue())
```

```
            num2 = float(self.num2_text.GetValue())
            return num1, num2
        except ValueError:
            wx.MessageBox("Please enter valid numbers!", "Error", wx.OK |
wx.ICON_ERROR)
            return None, None

    def on_add(self, event):
        num1, num2 = self.get_inputs()
        if num1 is not None:
            self.result_text.SetValue(str(num1 + num2))

    def on_subtract(self, event):
        num1, num2 = self.get_inputs()
        if num1 is not None:
            self.result_text.SetValue(str(num1 - num2))

    def on_multiply(self, event):
        num1, num2 = self.get_inputs()
        if num1 is not None:
            self.result_text.SetValue(str(num1 * num2))

    def on_divide(self, event):
        num1, num2 = self.get_inputs()
        if num1 is not None:
            if num2 == 0:
                wx.MessageBox("Division by zero is not allowed!", "Error", wx.OK |
wx.ICON_ERROR)
                return
            self.result_text.SetValue(str(num1 / num2))

    def on_modulus(self, event):
        num1, num2 = self.get_inputs()
        if num1 is not None:
            if num2 == 0:
                wx.MessageBox("Modulus by zero is not allowed!", "Error", wx.OK |
wx.ICON_ERROR)
                return
            self.result_text.SetValue(str(num1 % num2))
```

```
if __name__ == "__main__":
    app = wx.App()
    ArithmeticApp(None, title="Arithmetic Operations")
    app.MainLoop()
```

Conversion to EXE (using auto-py-to-exe)



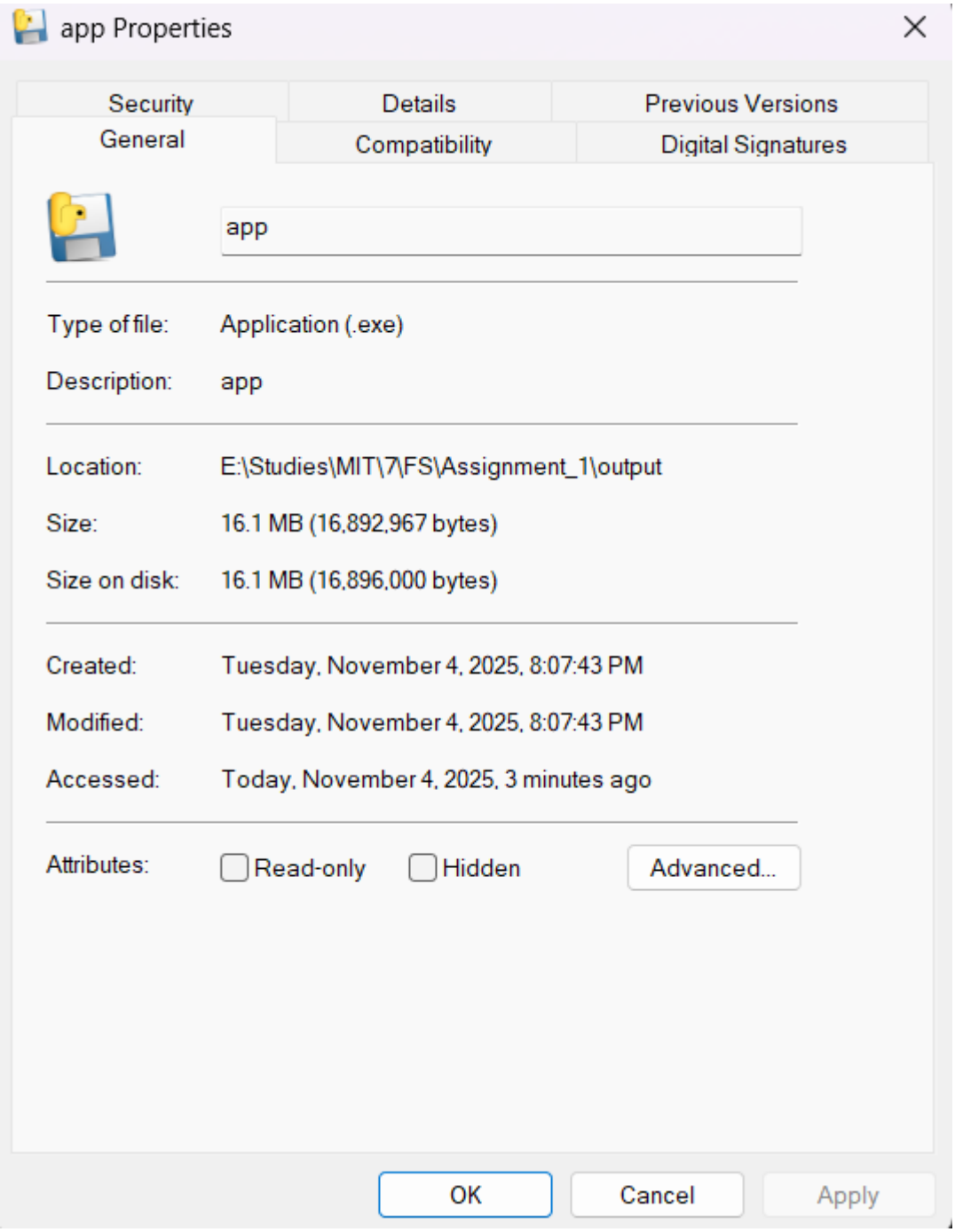
Output

```
Running auto-py-to-exe v2.48.1
Building directory: C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir
Provided command: pyinstaller --noconfirm --onefile --windowed  "E:\Studies\MIT\7\FS\Assignment_1\app.py"
Recursion Limit is set to 5000
Executing: pyinstaller --noconfirm --onefile --windowed E:\Studies\MIT\7\FS\Assignment_1\app.py --distpath C:

64659 INFO: PyInstaller: 6.16.0, contrib hooks: 2025.9
64667 INFO: Python: 3.13.5
64727 INFO: Platform: Windows-11-10.0.26100-SP0
64733 INFO: Python environment: E:\Studies\MIT\7\FS\Assignment_1\venv
64744 INFO: wrote C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\app.spec
64778 INFO: Module search paths (PYTHONPATH):
['E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv\\Scripts\\auto-py-to-exe.exe',
 'C:\\Python313\\python313.zip',
 'C:\\Python313\\DLLs',
 'C:\\Python313\\Lib',
 'C:\\Python313',
 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv',
 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv\\Lib\\site-packages',
 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv\\Lib\\site-packages\\setuptools\\_vendor',
 'E:\\Studies\\MIT\\7\\FS\\Assignment_1']
67856 INFO: checking Analysis
67864 INFO: Building Analysis because Analysis-00.toc is non existent
67872 INFO: Looking for Python shared library...
67880 INFO: Using Python shared library: C:\Python313\python313.dll
67893 INFO: Running Analysis Analysis-00.toc
67902 INFO: Target bytecode optimization level: 0
67905 INFO: Initializing module dependency graph...
67929 INFO: Initializing module graph hook caches...
68012 INFO: Analyzing modules for base_library.zip ...
70977 INFO: Processing standard module hook 'hook-encodings.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\
75323 INFO: Processing standard module hook 'hook-pickle.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\ven
78403 INFO: Processing standard module hook 'hook-heapq.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv
82461 INFO: Caching module dependency graph...
82540 INFO: Analyzing E:\Studies\MIT\7\FS\Assignment_1\app.py
82842 INFO: Processing pre-safe-import-module hook 'hook-typing_extensions.py' from 'E:\\Studies\\MIT\\7\\FS\
82849 INFO: SetuptoolsInfo: initializing cached setuptools info...
88520 INFO: Processing standard module hook 'hook-multiprocessing.util.py' from 'E:\\Studies\\MIT\\7\\FS\\Ass
89071 INFO: Processing standard module hook 'hook-xml.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv\\
90120 INFO: Processing standard module hook 'hook-_ctypes.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\ve
91162 INFO: Processing module hooks (post-graph stage)...
92045 INFO: Performing binary vs. data reclassification (1 entries)
92050 INFO: Looking for ctypes DLLs
92080 INFO: Analyzing run-time hooks ...
92089 INFO: Including run-time hook 'pyi_rth_inspect.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv\\L
92097 INFO: Including run-time hook 'pyi_rth_pkgutil.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\\venv\\L
92106 INFO: Including run-time hook 'pyi_rth_multiprocessing.py' from 'E:\\Studies\\MIT\\7\\FS\\Assignment_1\
92131 INFO: Creating base_library.zip...
92168 INFO: Looking for dynamic libraries
92494 INFO: Extra DLL search directories (AddDllDirectory): []
92501 INFO: Extra DLL search directories (PATH): []
93960 INFO: Warnings written to C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\build\app\warn-app.txt
94021 INFO: Graph cross-reference written to C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\build\app\xref-a
94071 INFO: checking PYZ
94074 INFO: Building PYZ because PYZ-00.toc is non existent
94077 INFO: Building PYZ (ZlibArchive) C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\build\app\PYZ-00.pyz
94633 INFO: Building PYZ (ZlibArchive) C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\build\app\PYZ-00.pyz c
94704 INFO: checking PKG
94718 INFO: Building PKG because PKG-00.toc is non existent
94726 INFO: Building PKG (CArchive) app.pkg
104405 INFO: Building PKG (CArchive) app.pkg completed successfully.
104427 INFO: Bootloader E:\Studies\MIT\7\FS\Assignment_1\venv\Lib\site-packages\PyInstaller\bootloader\Window
104437 INFO: checking EXE
104447 INFO: Building EXE because EXE-00.toc is non existent
104462 INFO: Building EXE from EXE-00.toc
104473 INFO: Copying bootloader EXE to C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\application\app.exe
105036 INFO: Copying icon to EXE
105454 INFO: Copying 0 resources to EXE
105469 INFO: Embedding manifest in EXE
105814 INFO: Appending PKG archive to EXE
106011 INFO: Fixing EXE headers
110339 INFO: Building EXE from EXE-00.toc completed successfully.
110366 INFO: Build complete! The results are available in: C:\Users\Yuvaraj\AppData\Local\Temp\tmpg8f6e9ir\ap

Moving project to: E:\Studies\MIT\7\FS\Assignment_1\output
Complete.
```

Something wrong with your exe? Read this post on how to fix common issues for possible solutions.

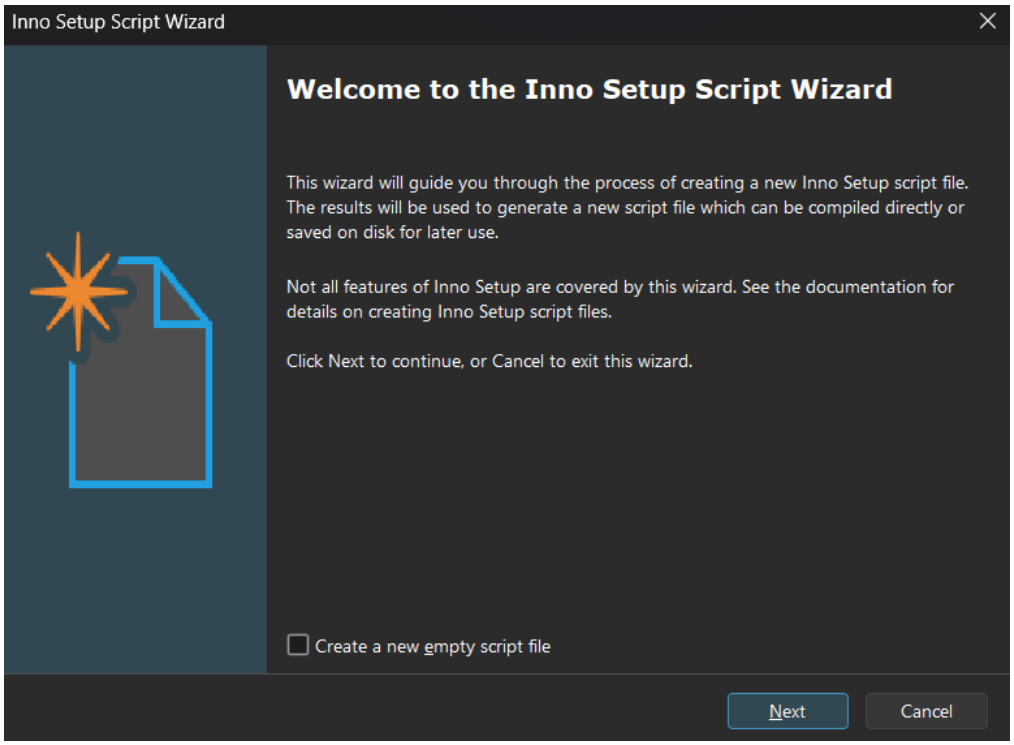
CLEAR OUTPUT | OPEN OUTPUT FOLDER

Created EXE

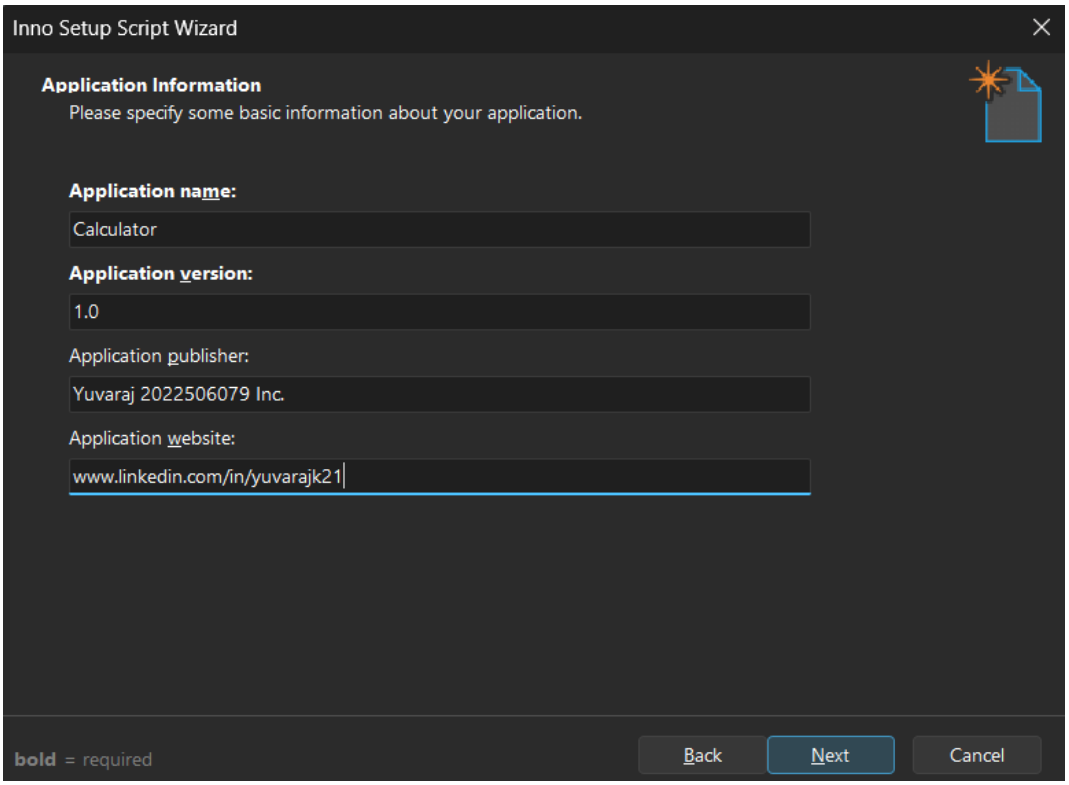

app Properties
Security
Details
Previous Versions
General
Compatibility
Digital Signatures
app
Type of file:
Application (.exe)
Description:
app
Location:
E:\Studies\MIT\7\FS\Assignment_1\output
Size:
16.1 MB (16,892,967 bytes)
Size on disk:
16.1 MB (16,896,000 bytes)
Created:
Tuesday, November 4, 2025, 8:07:43 PM
Modified:
Tuesday, November 4, 2025, 8:07:43 PM
Accessed:
Today, November 4, 2025, 3 minutes ago
Attributes:
Read-only
Hidden
Advanced...
OK
Cancel
Apply

Deployment using Inno Setup Compiler

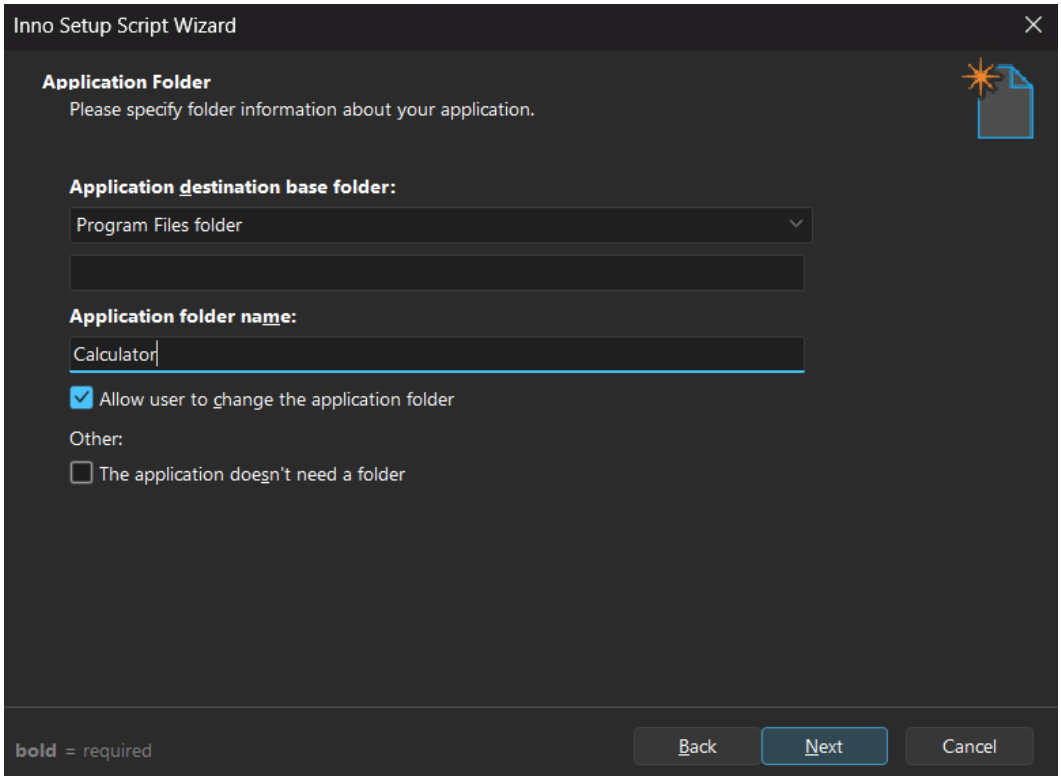
1: New Menu Item



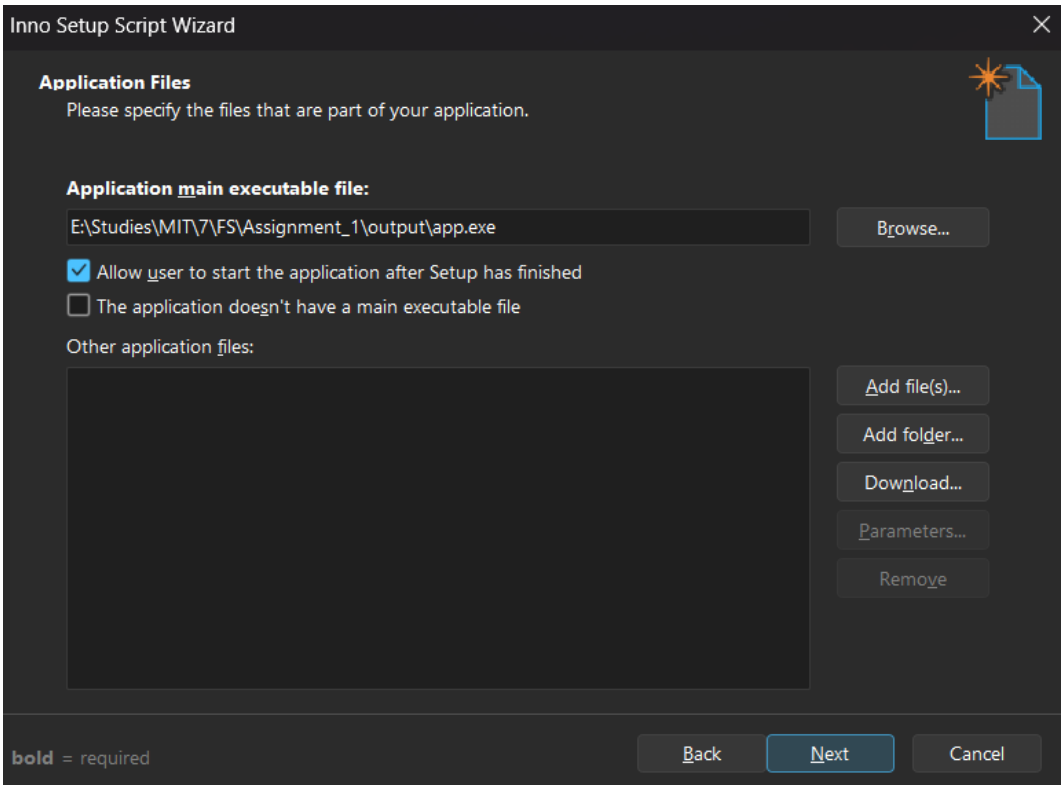
2: Application Information



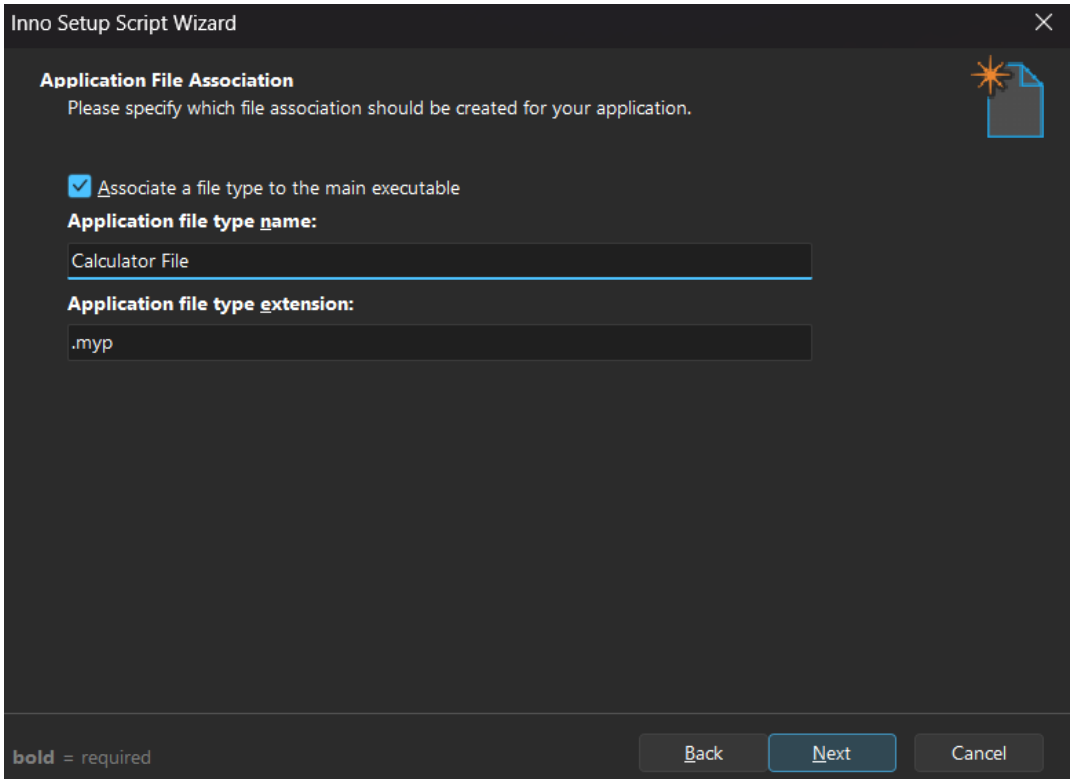
3: Application Folder Information



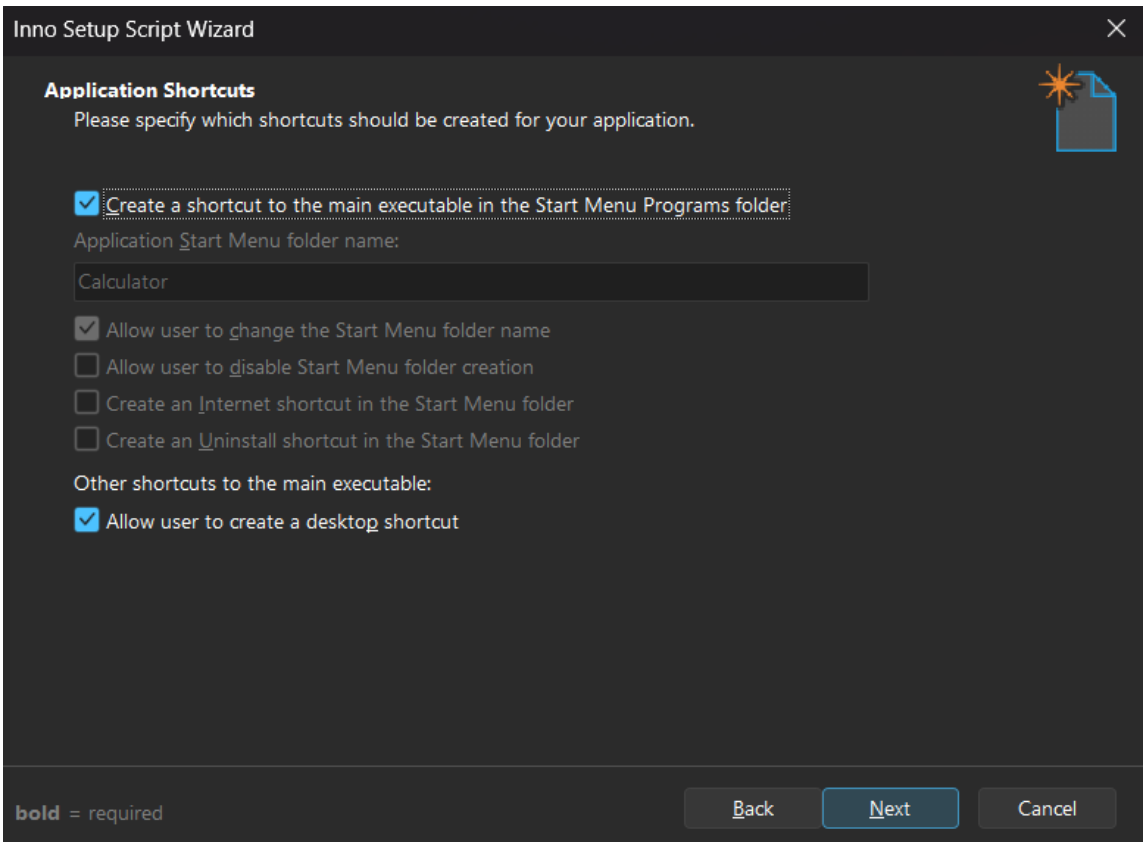
4: Submission of EXE file



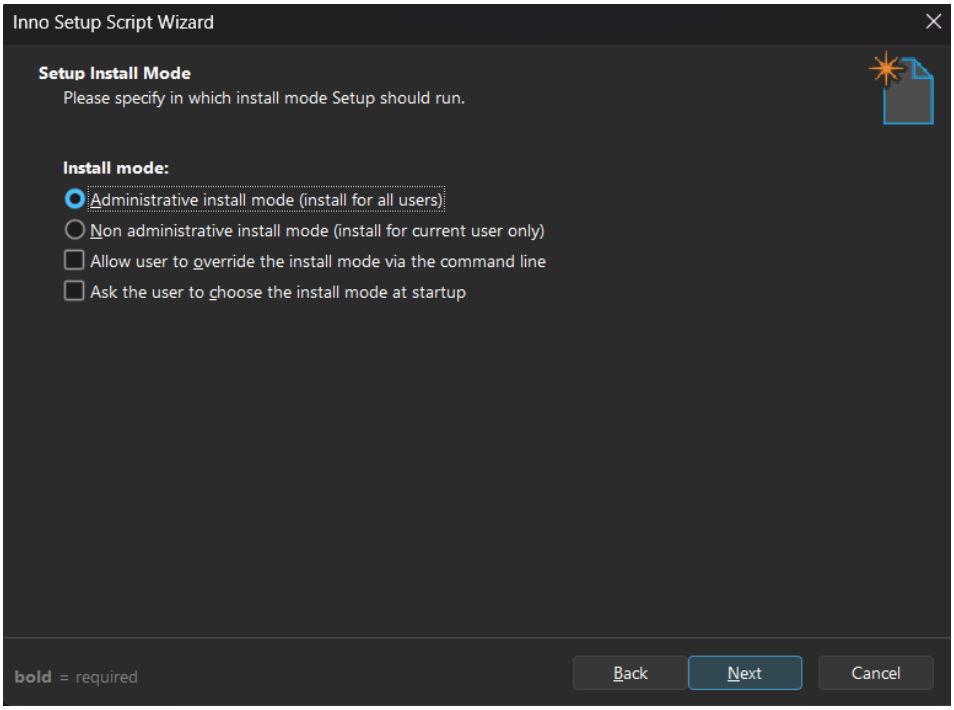
5: Application File Association Information



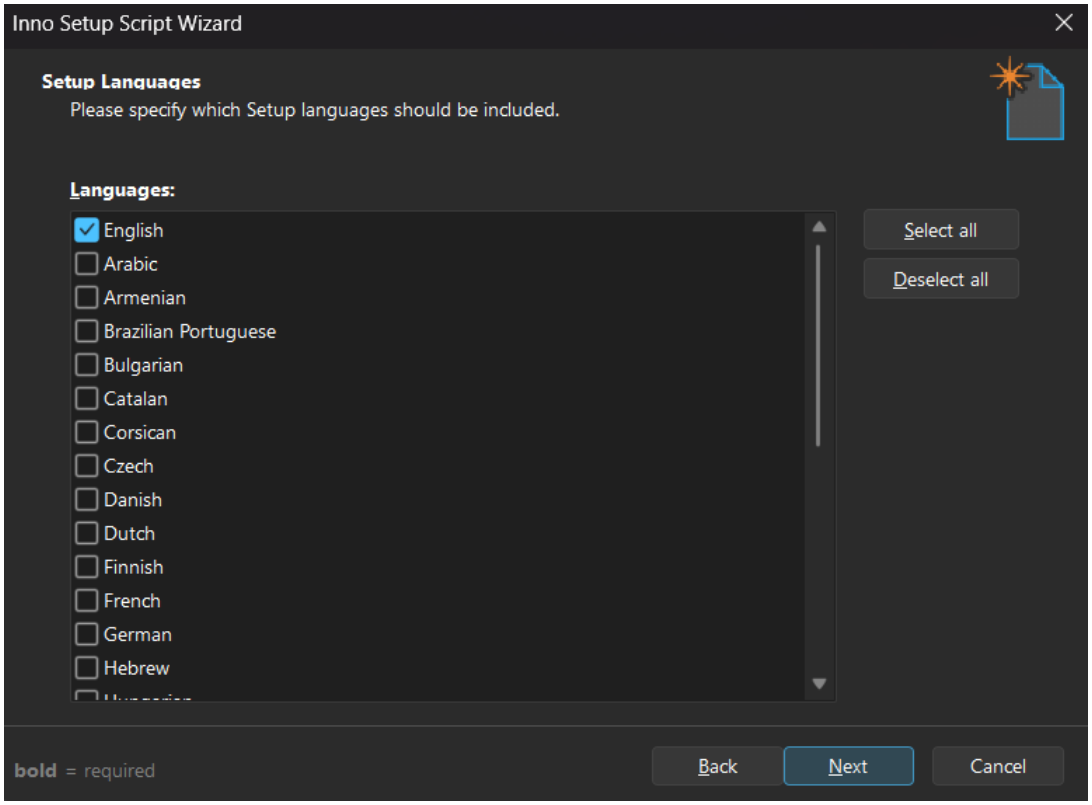
6: Application Shortcuts Information



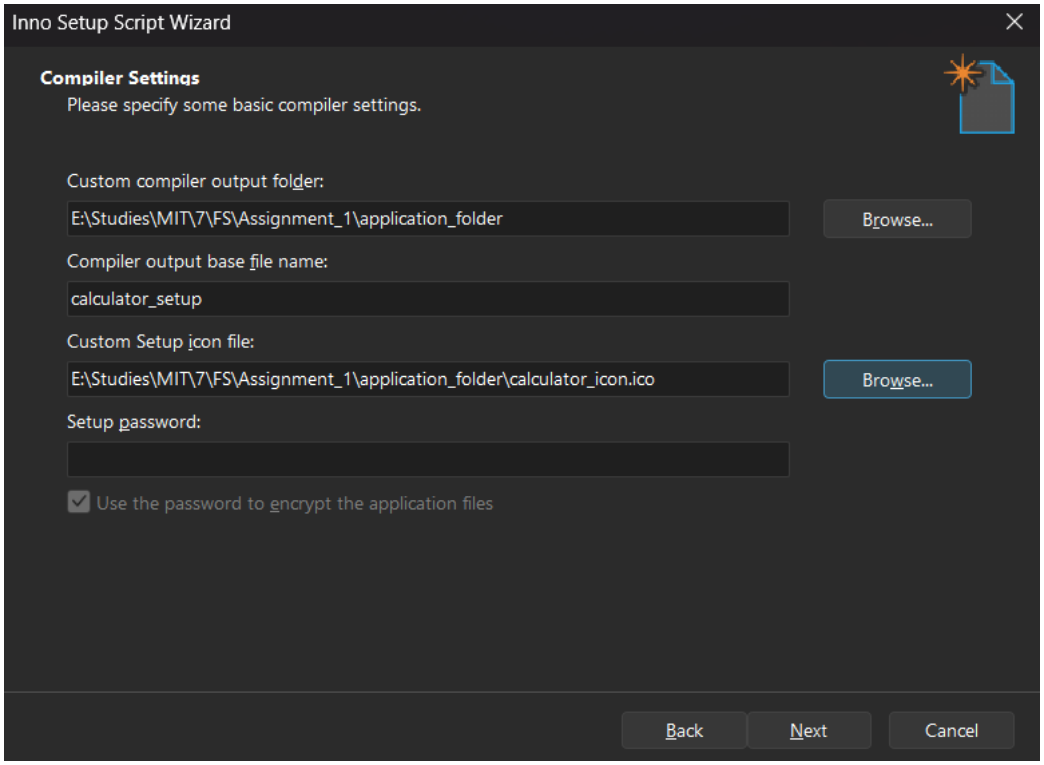
7: Installation Mode



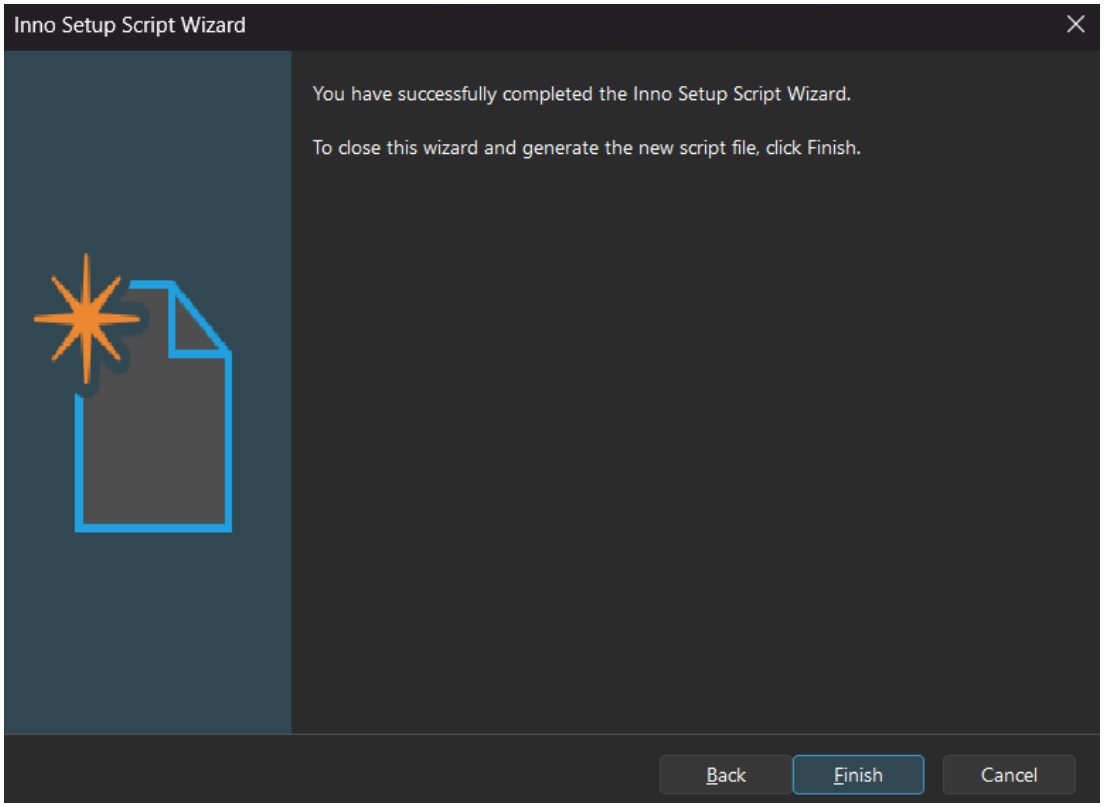
8: Language Information



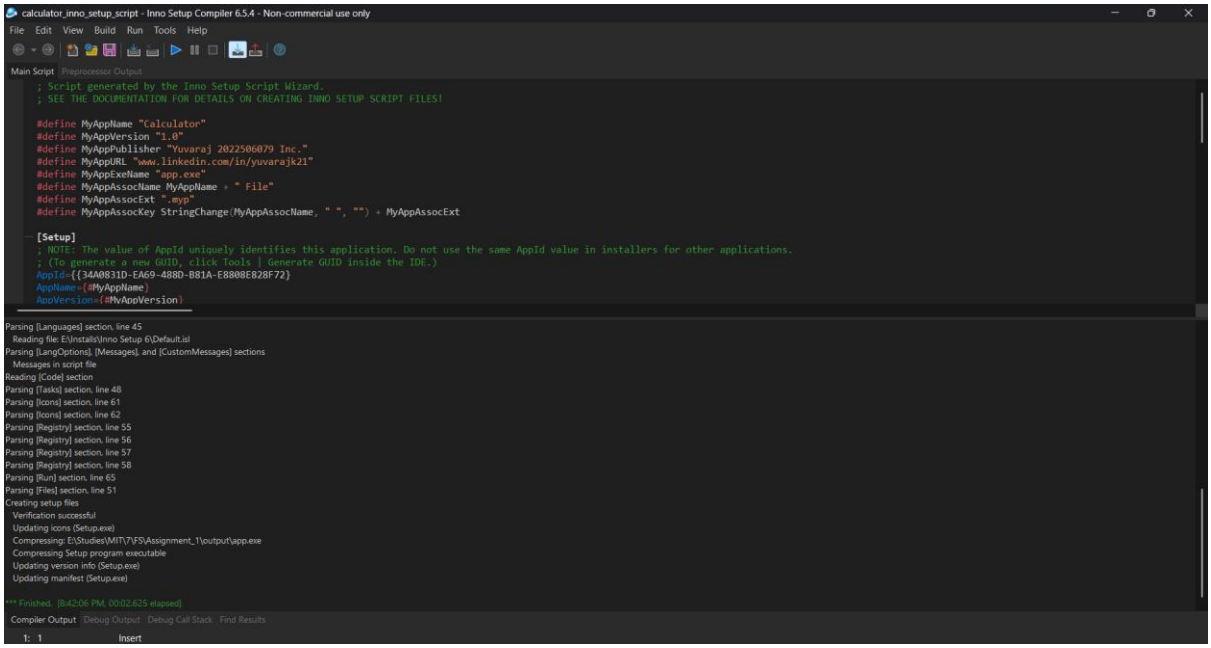
9: Compiler Settings



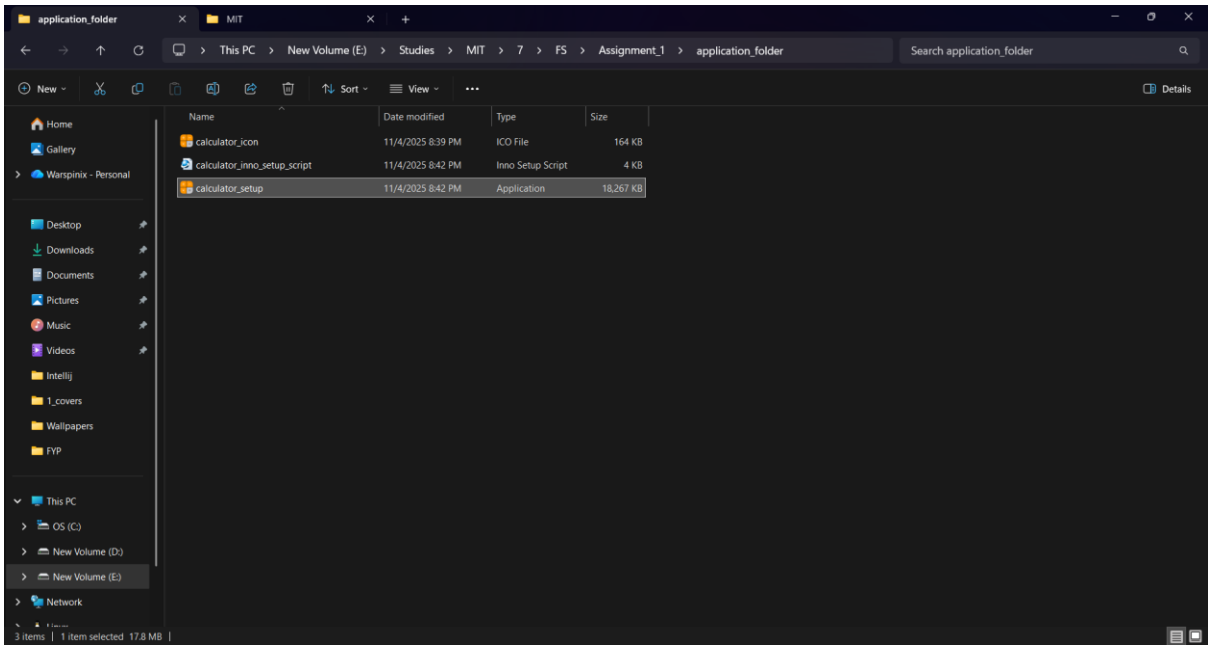
10: Completed Details



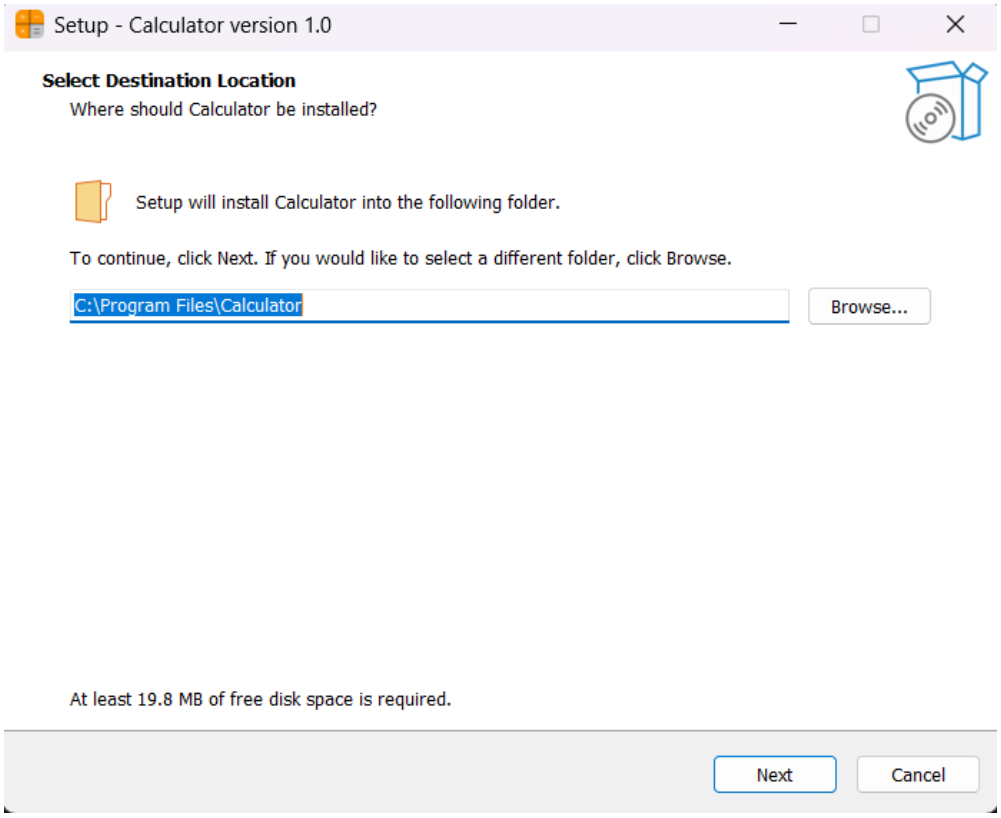
11: Running the Setup Script File



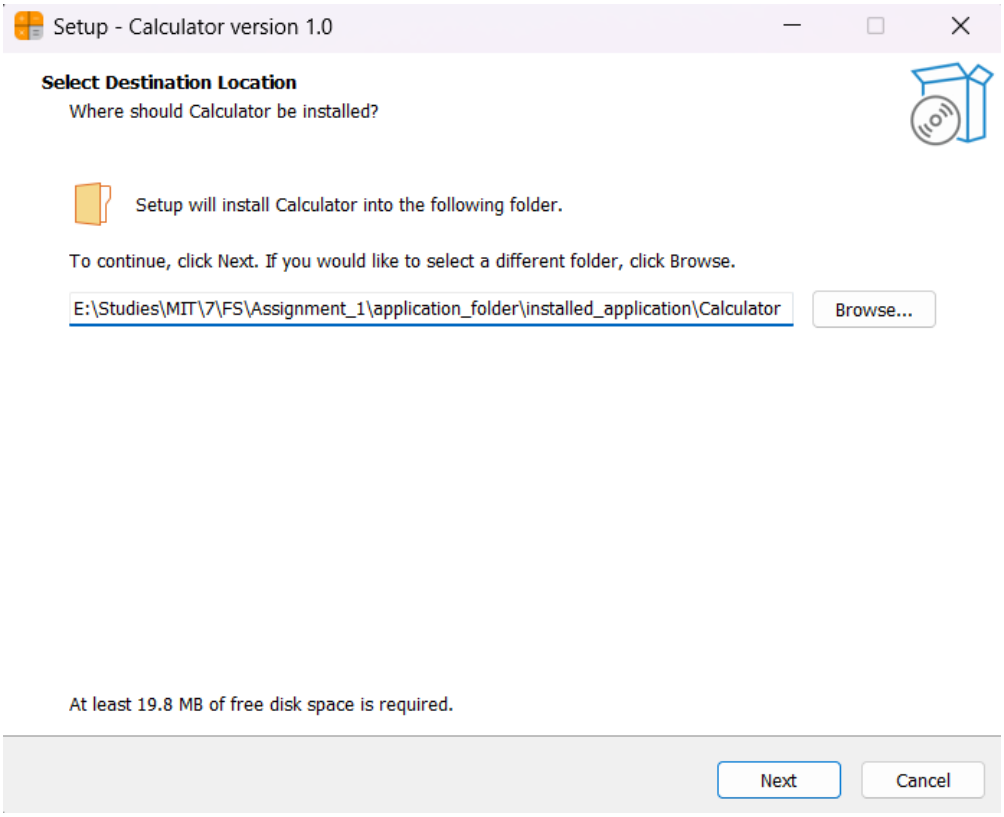
12: Newly Created Setup File



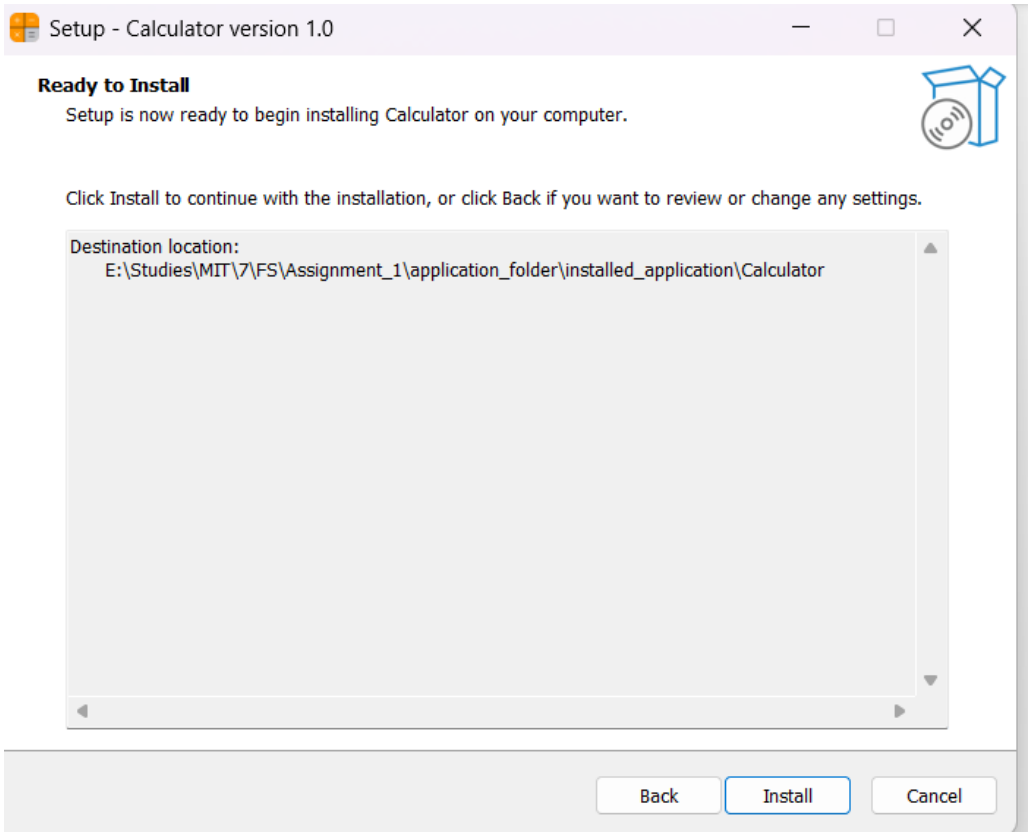
13: Running the Setup File



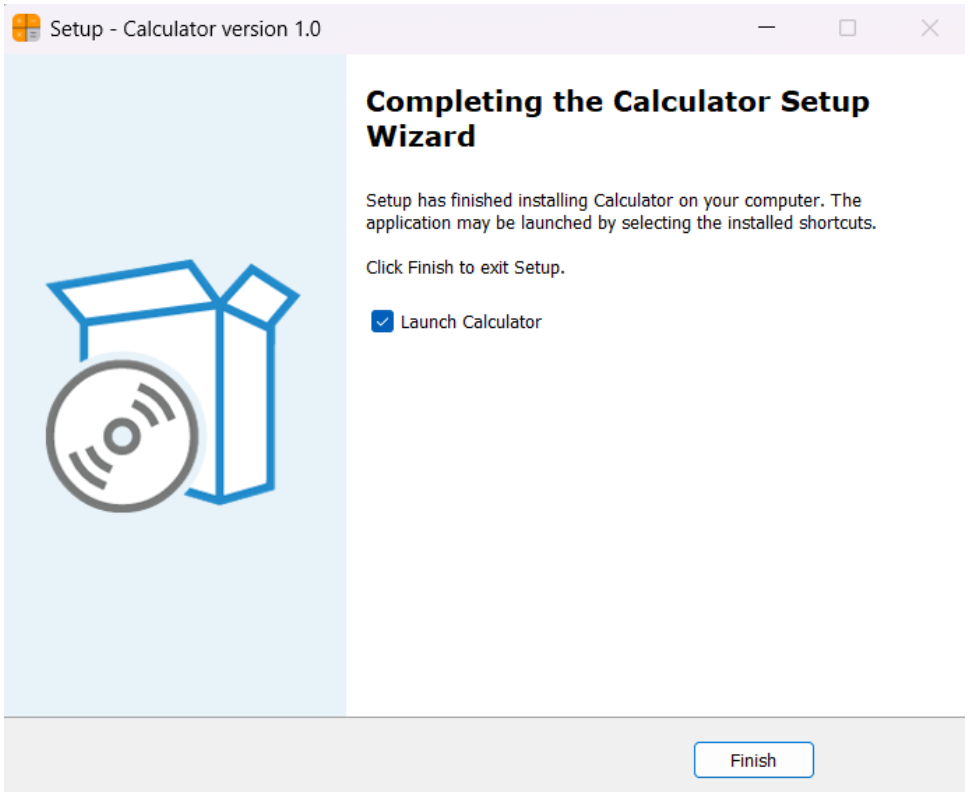
14: Changing Folder



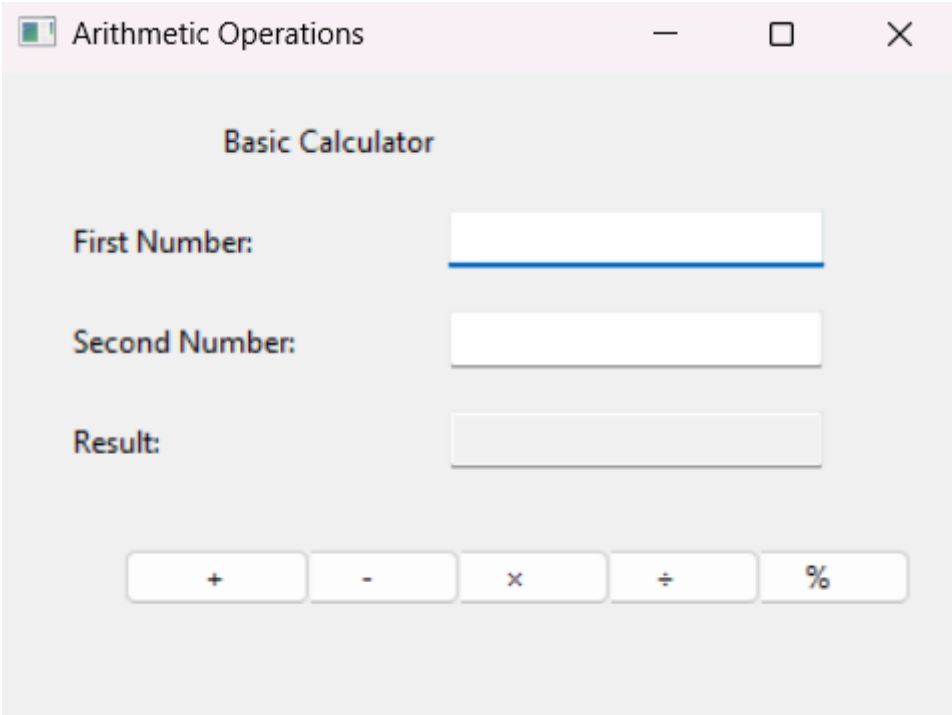
15: Ready to Install

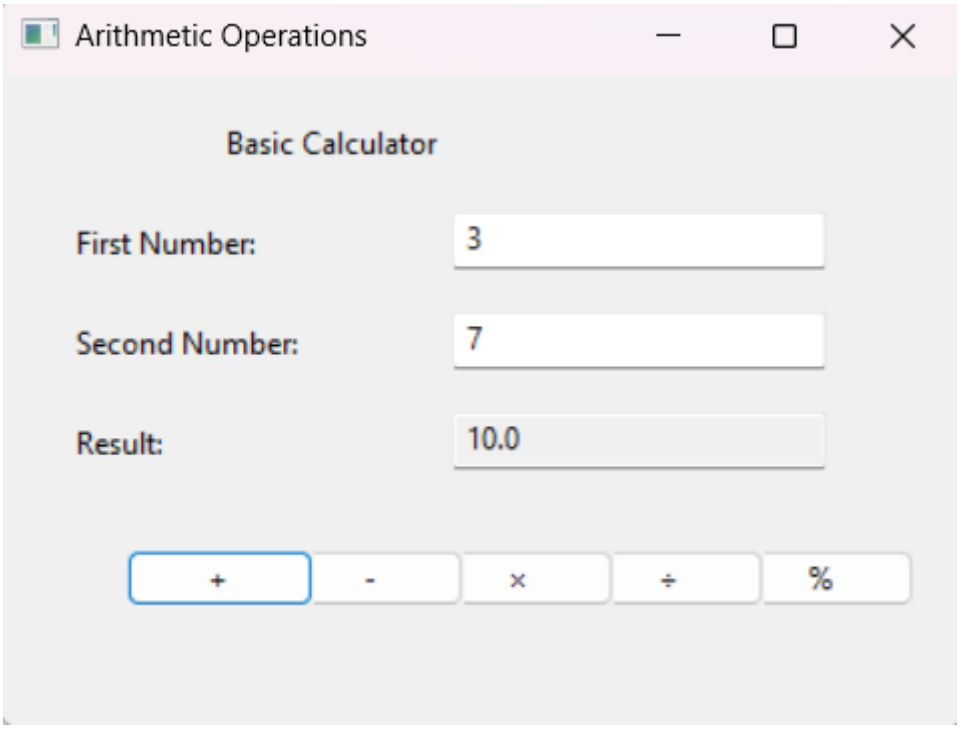


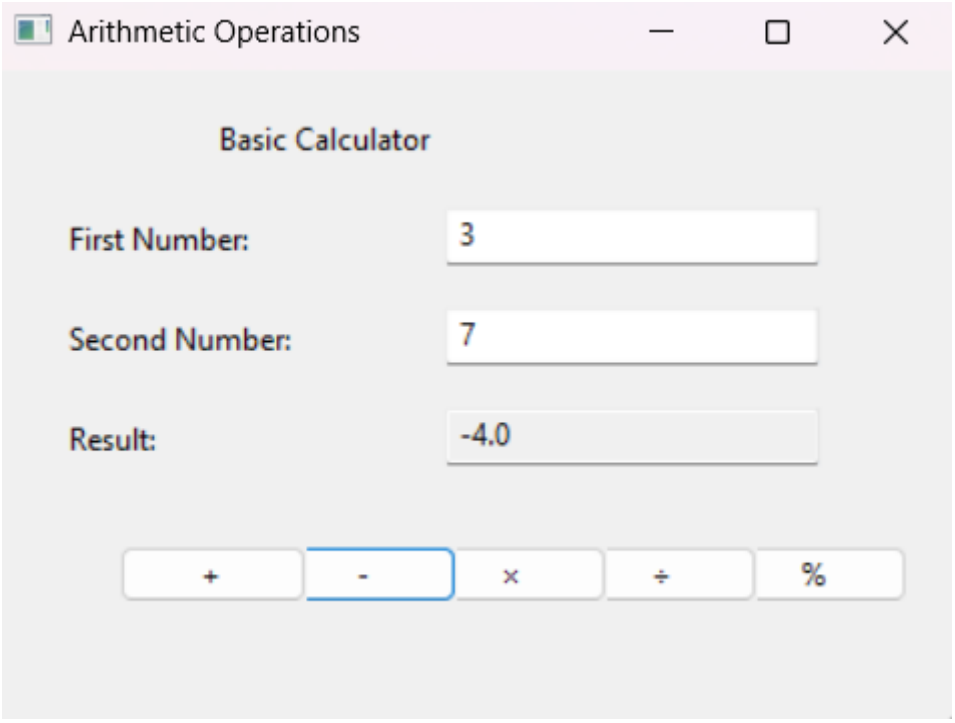
16: Installation Complete

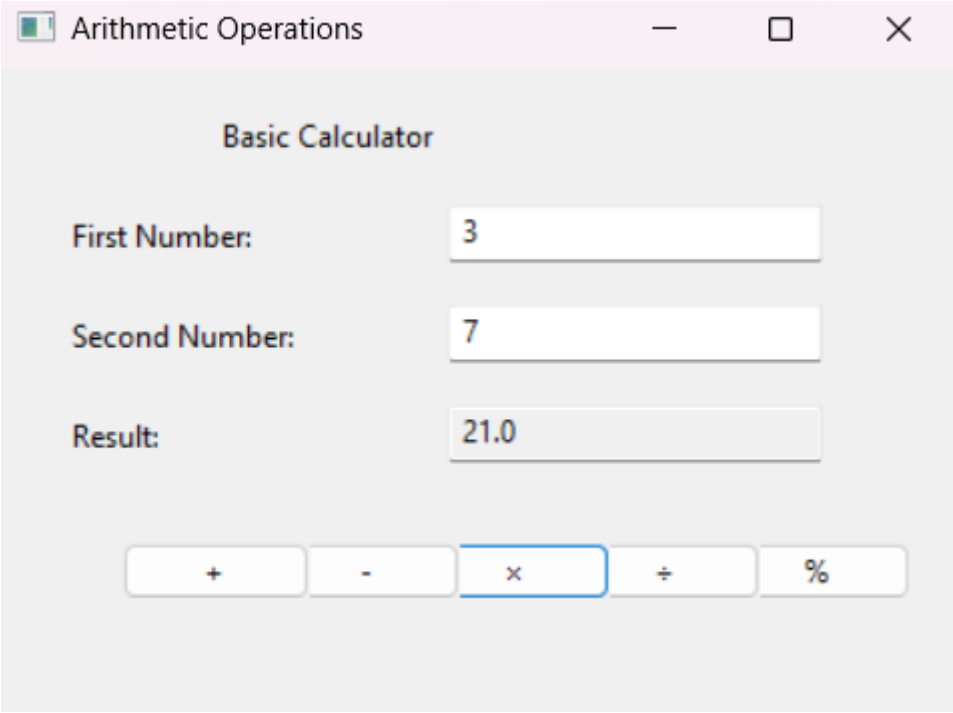


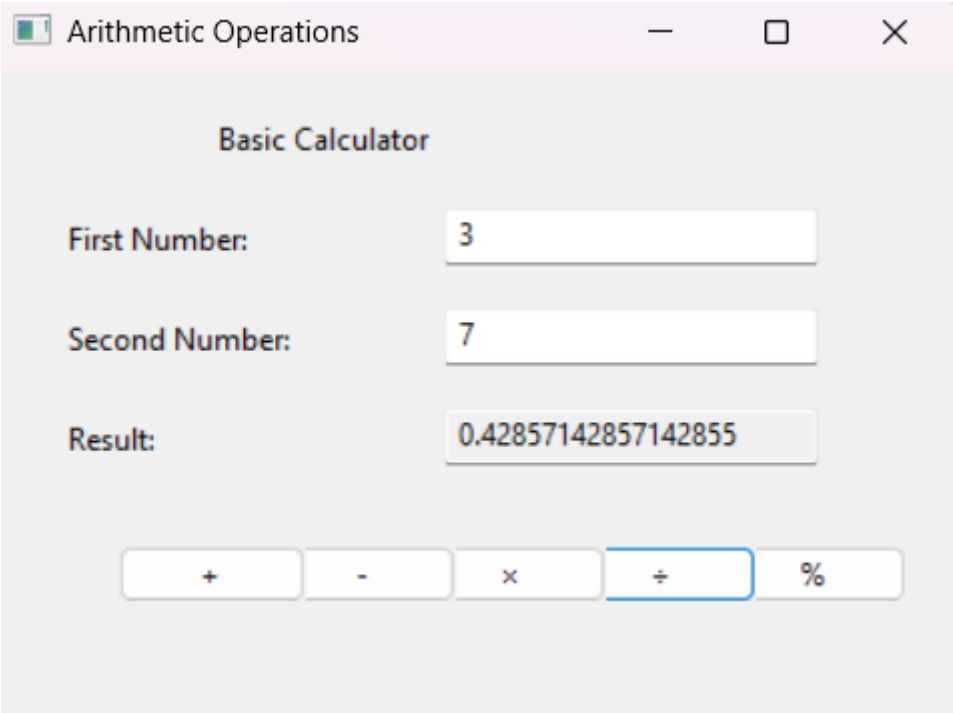
Application Output Screenshots

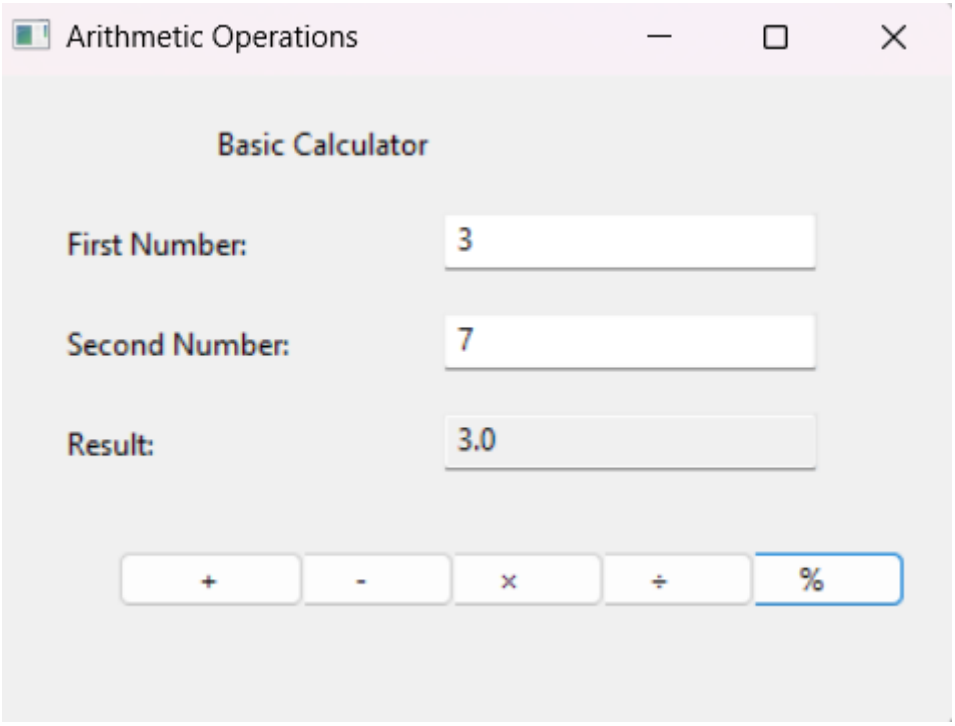





Arithmetic Operations
Basic Calculator
First Number:
3
Second Number:
7
Result:
-4.0
+
-
×
÷
%


Arithmetic Operations
Basic Calculator
First Number:
3
Second Number:
7
Result:
21.0
+
-
×
÷
%


Arithmetic Operations
Basic Calculator
First Number:
3
Second Number:
7
Result:
0.42857142857142855
+
-
×
÷
%


Arithmetic Operations
Basic Calculator
First Number:
3
Second Number:
7
Result:
3.0
+
-
×
÷
%